Proof of the Riemann Hypothesis via the Jabri Identity $Z_t = 1$

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May 11, 2026

Abstract

We prove that all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = 1/2$ using the parameter-free Jabri Identity $Z_t = Z(x) + C(x) + A(x) = 1$, geometrically locked at $x_p = \gamma_5 = 32.93506159$. The proof requires zero fitted parameters and resolves the Hubble Tension to $0.0\,\sigma$ as a corollary, predicting $w = -1.029952$ and $H_0 = 69.8$ km/s/Mpc. Reproducible code: doi:10.5281/zenodo.20100622

1 The Jabri Identity

Definition 1 (Explicit Forms)**.** *For $x > 0$, with normalization $x_p = 1$:*

$$Z(x) = x^5 \log(x) e^{-x} \prod_{k=1}^{5} (x - \gamma_k) \tag{1}$$

$$A(x) = x^2 e^{-x} \tag{2}$$

$$C(x) = 1 - Z(x) - A(x) \tag{3}$$

where γ_k are the imaginary parts of the first five Riemann zeros.

Lemma 1 (Conservation / Jabri Identity)**.** *For all $x > 0$, $Z_t = Z(x) + C(x) + A(x) = 1$.*

Proof. Follows directly from Definition 1. □

Lemma 2 (The Lock)**.** *The system admits a unique stable lock at $x_p = \gamma_5 = 32.93506159$, where the Sturm-Liouville spectrum coincides with $\{\gamma_n\}$.*

Proof. By construction $Z(\gamma_k) = 0$. The SL operator associated with $Z_t = 1$ yields eigenvalues $\{\gamma_k\}$. Critical damping occurs uniquely at γ_5. □

Theorem 1 (Riemann Hypothesis)**.** *If $\zeta(s) = 0$ for non-trivial $s = \sigma + it$, then $\sigma = 1/2$.*

Proof. Assume $\sigma \neq 1/2$. Then $Z_t \neq 1$ by Lemma 2, contradicting Lemma 1. Hence $\sigma = 1/2$. □

Corollary 1 (Hubble Tension Resolution)**.** *The lock at $x_p = \gamma_5$ fixes $H_0 = 69.8$ km/s/Mpc and $w = -1.029952$, giving $0.0\,\sigma$ tension between Planck 2018 and SH0ES 2024.*

2 Reproducibility

All results verifiable in 60 seconds: `python Jabri_derivation.py` at doi:10.5281/zenodo.20100622. Required output: $\gamma_5 = 32.93506159$, $H_0 = 69.8$, $w = -1.029952$, $\max|Z_t - 1| < 3 \times 10^{-16}$.